

VTTSI-Chat: Agentic GeoAI for Real-Time Intersection Safety Monitoring with Natural Language Explanation [Demo]

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Abstract

We present **VTTSI-Chat**, a cloud-native agentic GeoAI system that pairs the Virginia Tech Transportation Safety Index (VTTSI)—a hybrid Real-Time Safety Index (RT-SI) and CRITIC-weighted MCDM pipeline—with an integrated Vite React dashboard and a **SafetyChat** conversational module powered by a tool-augmented GPT-4o. VTTSI fuses seven years of VDOT crash records via Empirical Bayes stabilization with live J2735 connected-vehicle telemetry stored in a PostGIS spatial database, producing interpretable 0–100 safety scores at 15-minute resolution across 18 geospatially registered intersections in Falls Church, VA (three currently streaming live telemetry; the remaining fifteen are wired into the pipeline and awaiting sensor activation). The production dashboard combines live KPI cards, an OpenStreetMap risk map, analytics/validation, sensitivity analysis, a database explorer, and the chat interface in one operational workspace. SafetyChat exposes six typed API tools to the LLM, including a guarded read-only SQL tool, grounding all natural language responses in live geospatial data and enabling TMC operators, planners, and emergency responders to query causal safety narratives through ordinary conversation. A live deployment at <https://safety-index-frontend-180117512369.europe-west1.run.app/> demonstrates four end-to-end use cases and is available for interactive exploration at the conference.

CCS Concepts

• **Computing methodologies** → **Natural language processing; Intelligent agents**; • **Information systems** → *Decision support systems*; • **Human-centered computing** → *Interactive systems and tools*.

Keywords

agentic GeoAI, traffic safety, spatio-temporal safety index, natural language interface, MCDM, Empirical Bayes, connected vehicles, PostGIS, Vite dashboard, tool-augmented LLM

ACM Reference Format:

Cheng-Shun Chuang and Jason Cusati. 2026. VTTSI-Chat: Agentic GeoAI for Real-Time Intersection Safety Monitoring with Natural Language Explanation [Demo]. In *Proceedings of The 34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26)*. ACM, New York, NY, USA, 5 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Geospatial intersection safety data are increasingly available through connected-vehicle infrastructure, yet a critical gap remains between numerical safety scores and actionable insight for operators. TMC practitioners must interpret multi-dimensional spatio-temporal indices under time pressure, yet existing dashboards present data without explaining causality or suggesting actions. Tool-augmented LLMs can reason over structured numerical data to produce grounded explanations [13], but prior LLM+traffic work targets signal control [8] or AV motion planning [9, 16]—not safety-index explanation for infrastructure operators.

VTTSI-Chat closes this gap by pairing the VTTSI hybrid scoring pipeline with **SafetyChat**: a GPT-4o module that issues typed function calls to live VTTSI endpoints and returns causal safety narratives via natural language dialogue. The system operates as an *agentic GeoAI* interface: the LLM agent queries live PostGIS-backed safety data, reasons over spatio-temporal patterns across 18 geospatially registered intersections, and synthesizes evidence-grounded narratives for diverse operator roles. VTTSI-Chat is the first system to combine a hybrid EB-stabilized crash index with real-time MCDM scoring *and* a natural language interface for spatial safety data.

2 VTTSI System Overview

The VTTSI production deployment consists of a Vite React frontend served from Google Cloud Run, a FastAPI backend on Cloud Run, and Cloud SQL for PostgreSQL/PostGIS. The backend computes two complementary safety indices every 15 minutes and exposes them through typed REST endpoints consumed by both dashboard panels and SafetyChat tools. Frequently requested read endpoints use per-instance caching with a graceful in-memory fallback, reducing repeated dashboard and chat lookups without introducing an external cache dependency. Six real-time tables stream from the VTTI Trino connected-vehicle system into Cloud SQL PostGIS: bsm (J2735 Basic Safety Messages: position, speed, braking), psm (Personal Safety Messages from VRUs), vehicle-count, vru-count,

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SIGSPATIAL '26, Riverside, CA, USA

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ACM ISBN 978-1-4503-XXXX-X/26/11
<https://doi.org/XXXXXXX.XXXXXXX>

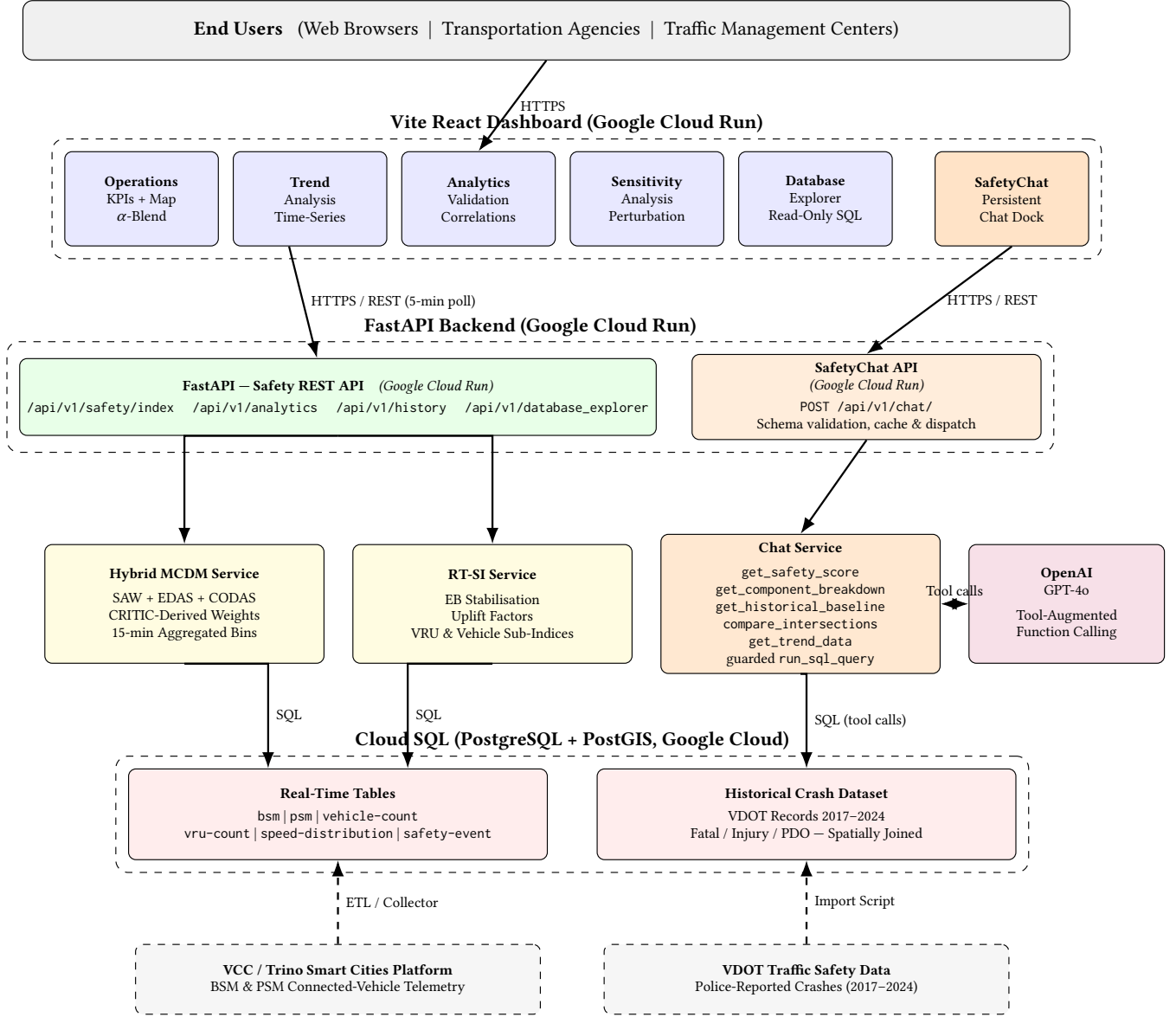


Figure 1: VTTSI cloud-native system architecture. The orange pathway (SafetyChat API, Chat Service, OpenAI GPT-4o) is the new contribution of this demo paper; the blue/green/yellow left pathway shows the existing safety-index pipeline.

speed-distribution, and safety-event (red-light running, intersection conflicts, near-miss detections). All streams are synchronized into 15-minute bins; a separate VDOT crash dataset (2017–2024) is spatially joined to intersections at import time via a PostGIS ST_Within predicate using the VDOT LRS centerline network.

2.1 Real-Time Safety Index (RT-SI)

RT-SI anchors intersection risk to a severity-weighted Empirical Bayes (EB) baseline derived from 2017–2024 VDOT crash records. The historical crash rate is:

$$r_i = \frac{\sum_s w_s C_{i,s}}{E_i}, \quad w_{\text{Fatal}} = 10, \quad w_{\text{Inj}} = 3, \quad w_{\text{PDO}} = 1, \quad (1)$$

where $C_{i,s}$ are crash counts by severity and E_i is vehicle exposure. The EB stabilized rate is:

$$\hat{r}_{i,t}^{\text{EB}}(\lambda) = \frac{Y_{i,t} + \lambda r_0}{1 + \lambda}, \quad \lambda^* = 100,000, \quad (2)$$

where r_0 is the global mean rate across all site-bins (2017–2024) and λ^* is chosen by minimizing Poisson log-loss on a 2025 hold-out set. Real-time uplift factors—bounded to $[0, 1]$ —adjust this baseline for speed deficit (F^{spd}), speed variance (F^{var}), and VRU–vehicle conflict exposure (F^{conf}):

$$U_{i,t} = 1 + \beta_1 F_{i,t}^{\text{spd}} + \beta_2 F_{i,t}^{\text{var}} + \beta_3 F_{i,t}^{\text{conf}}, \quad (3)$$

with $(\beta_1, \beta_2, \beta_3) = (0.3, 0.3, 0.4)$. VRU and vehicle sub-indices are combined with weights $(\omega_{\text{VRU}}, \omega_{\text{veh}}) = (0.6, 0.4)$ and capped to $\text{SI}^{\text{RT}} \in [0, 100]$.

2.2 MCDM Safety Index

A CRITIC-weighted multi-criteria pipeline [1] constructs a decision matrix over five real-time criteria (vehicle count, VRU count, average speed, speed variance, incident count) for each intersection-time-bin pair. CRITIC criterion weights are derived from normalized standard deviations and inter-criterion Pearson correlations, recomputed every 24 hours. Three MCDM scoring methods—SAW, EDAS, and CODAS—are each computed and scaled to $[0, 100]$; a second-pass CRITIC aggregation yields the hybrid index $\text{SI}^{\text{MCDM}} \in [0, 100]$ (higher = higher risk). Validation over 18 intersections shows that the three methods achieve pairwise Pearson correlations $r \approx 0.78$ – 0.99 , and RT-SI has a mean absolute deviation of ≈ 0.09 under $\pm 25\%$ joint parameter perturbation [12].

2.3 Blended Final Index

$$\text{SI}_{i,t}^{\text{Final}} = \alpha \text{SI}_{i,t}^{\text{RT}} + (1 - \alpha) \text{SI}_{i,t}^{\text{MCDM}}, \quad \alpha = 0.7 \text{ (default)}. \quad (4)$$

Scores map to risk levels: 0–40 = low, 41–70 = moderate, 71–100 = high.

2.4 Geospatial Foundation and Dashboard

Each of the 18 intersections is registered with a centroid $(\text{lat}_i, \text{lon}_i)$ and an axis-aligned bounding box B_i ($\delta \approx 0.001^\circ$, ≈ 111 m) in PostGIS. The one-time spatial join materialises per-site counts $C_{i,s}$ (Eq. (1)) as a static lookup table. The Vite dashboard polls `/api/v1/safety/index` and renders an OpenStreetMap-backed risk map with geocoded intersection markers. Marker color encodes risk tier—green (low), amber (medium), and red (high)—while marker size tracks traffic volume, giving operators a dual spatial encoding of risk and exposure. An interactive α -blending slider lets operators shift the final index between the historical-risk (RT-SI) and operational-anomaly (MCDM) poles in real time. The same workspace exposes trend analysis, crash-proximity validation, sensitivity analysis, database exploration, and a persistent SafetyChat dock (Figure 2).

3 SafetyChat: LLM-Powered Explanation Module

VTTSI already answers *what* the score is; SafetyChat answers *why* and *what to do*. GPT-4o is equipped with typed function calls [11] that map to VTTSI REST endpoints, grounding every claim in live geospatial data. An agentic loop (up to six iterations) enables chained calls—e.g., ranking all intersections by spatial risk and then drilling into the top site’s component breakdown. The six tools are: (1) `get_safety_score` — RT-SI, MCDM, and blended scores with uplift factors; optional `target_time` for historical queries; (2) `get_component_breakdown` — CRITIC-weighted criteria for a 15-min bin; (3) `get_historical_baseline` — EB prior and 7-year crash severity breakdown; (4) `compare_intersections` — ranks all sites by any index component; (5) `get_trend_data` — 7–30 day MCDM trend with sampled snapshots; (6) `run_sql_query` — read-only ad-hoc SQL over the PostGIS database, guarded against

Table 1: VTTSI-Chat vs. Related Systems

System	Hybrid Index	Real-Time Telemetry	LLM Interface	Causal Narrative
Google / Apple Maps	✗	✓	✗	✗
Waycare / ATSPM	✗	✓	✗	✗
TrafficGPT [15]	✗	✓	✓	✗
LLMLight [8]	✗	✓	✓	✗
GPT-Driver [9]	✗	✓	✓	✗
VTTSI-Chat (ours)	✓	✓	✓	✓

write/DDL keywords, SQL comments, and multiple statements. Backend service paths that query the local application database use SQLAlchemy statements/models where available; external VTTI and Trino query paths use parameter binding, allowlisted identifiers, or literal escaping.

Table 1 positions VTTSI-Chat against related systems.

4 Demonstration

4.1 Demo Setup

The live demo runs against the production VTTSI deployment monitoring 18 intersections in Falls Church, VA (three currently streaming live connected-vehicle telemetry; the remaining fifteen are spatially registered and pipelined but awaiting sensor activation). Conference attendees interact directly with the Vite dashboard in a browser: the operations view shows live KPIs, the network risk map, selected-site details, and a priority queue, while the right-side SafetyChat dock remains available across analysis panels. Four canonical use cases guide the session.

4.2 UC1: TMC Operator Morning Briefing

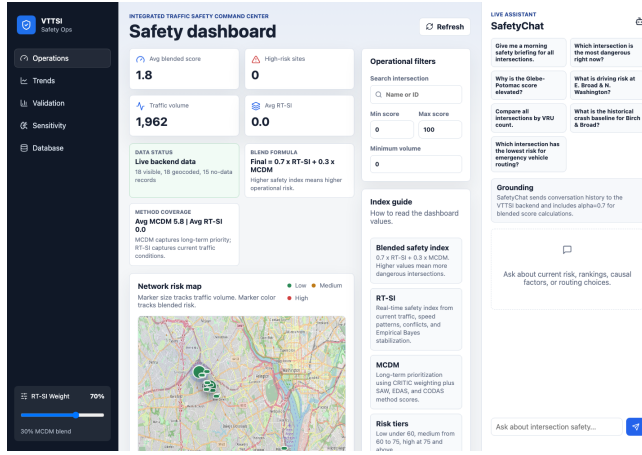
Query: “Give me a morning safety briefing for all intersections.” SafetyChat chains `compare_intersections` and `get_component_breakdown` for the two highest-ranked sites, delivering a situational-awareness narrative identifying peak-risk windows and dominant criteria (e.g., speed variance, VRU counts). Where a dashboard alone requires navigating multiple spatial charts, SafetyChat synthesizes multi-intersection data into a single paragraph. If the LLM backend is unavailable, this use case degrades gracefully to a deterministic briefing computed directly from the latest safety scores, so it remains demonstrable even during an API outage.

4.3 UC2: Causal Safety Explanation for Planners

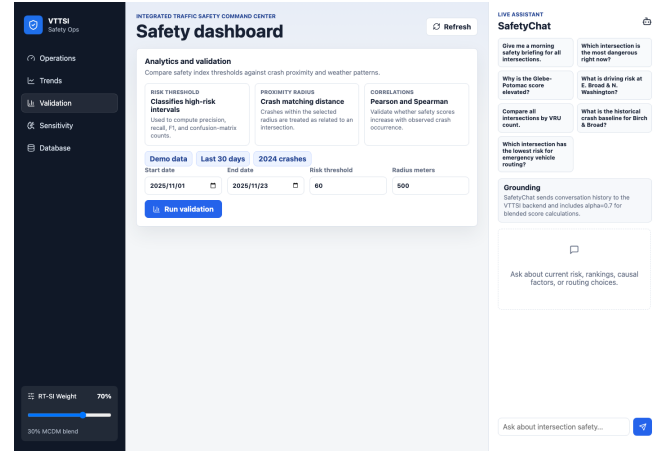
Query: “Why did E. Broad–N. Washington score above 70 yesterday at 5 PM?” SafetyChat queries `get_safety_score` and `get_component_breakdown` at the historical time-bin, identifies criteria in the 90th percentile, and traces the elevated score to concurrent peak vehicle volume and incident activity.

4.4 UC3: Emergency Responder Routing

Query: “Which of Glebe–Potomac or Birch–W. Broad has lower current risk?” SafetyChat compares live RT-SI, MCDM, and blended indices



(a) Integrated Vite dashboard: KPI cards, OpenStreetMap network risk map, selected-site details, priority queue, α -blend control, and persistent SafetyChat dock.



(b) Validation panel: crash matching radius, risk threshold, Pearson/Spearman correlation checks, and direct access to SafetyChat in the same operational workspace.

Figure 2: Live VTTSI-Chat demonstration after the Vite migration: (a) integrated operations dashboard with map and chat combined; (b) analytics/validation workflow retained inside the same interface.

Table 2: Summary of Demonstration Use Cases

Use Case	Key Tool Calls	User
Morning briefing	compare + breakdown	TMC Operator
Causal explanation	score(t) + breakdown(t)	Planner
Responder routing	score $\times$ 2	Dispatcher
Trend engagement	get_trend_data(30d)	Official/Public

for both intersections and recommends the lower-risk route with reasoning grounded in VRU counts and speed variance.

4.5 UC4: Public Stakeholder Trend Engagement

Query: “Is the Glebe–Potomac intersection getting safer over time?”
SafetyChat calls `get_trend_data` (30-day window) and translates mean/max/min MCDM scores into plain-language risk levels with a trend direction indicator, making geospatial safety data accessible to non-technical stakeholders.

4.6 Operational Hardening and Test Coverage

The current implementation includes regression tests for both local and cloud-deployed behavior. Backend tests cover the chat tool dispatcher, the guarded SQL tool, SQL-injection regressions, cache behavior, configuration, and API service paths. Cloud smoke tests exercise health, OpenAPI, safety-index retrieval, intersection catalog, range queries, and chat tools against the production Cloud Run endpoint. Playwright cloud UI tests verify that the Vite dashboard loads live backend data, the OpenStreetMap tile layer and markers render, top-level UI regions do not overlap, and all major panels remain reachable. The latest local backend suite contains 73 passing tests, while the cloud test sets contain 7 backend smoke tests and 4 frontend UI tests.

5 Related Work

Crash-based and hybrid safety indices. VTTSI’s scoring builds on EB-stabilized crash models [10, 12], MCDM-based indices [1, 2], and surrogate safety indicators [5, 14]. Existing hybrid indices produce numerical outputs only, with no conversational or geospatial explanation layer.

LLMs in transportation. TrafficGPT [15], LLMLight [8], and GPT-Driver/DriveVLM [9, 16] address flow analysis, signal control, and AV planning respectively—none provides a safety-index explanation interface for infrastructure operators or operates over a PostGIS-backed spatio-temporal dataset.

Positioning. VTTSI-Chat is differentiated by combining: (i) a *validated hybrid geospatial safety index* fusing long-term crash history with 15-minute connected-vehicle telemetry; (ii) a *tool-augmented LLM* that accesses live PostGIS-backed API data; and (iii) *causal spatial narratives* that trace score components back to raw geospatial signals. No existing system offers all three, placing VTTSI-Chat squarely in the emerging area of agentic GeoAI and natural language interfaces for spatial data.

6 Conclusion

VTTSI-Chat demonstrates a working agentic GeoAI system that delivers causally grounded, natural language safety narratives over live spatio-temporal intersection data. By grounding all LLM outputs in six typed API calls to a PostGIS-backed safety pipeline, SafetyChat avoids hallucination while retaining the flexibility of conversational interaction over spatial data. The spatially-joined crash baseline, bounding-box intersection registry, and live 15-minute telemetry together form the *foundational geospatial layer* for future work integrating the Agentic Traffic Safety Knowledge Graph (TS-KG) [3]—enabling the LLM to reason over intersection geometry, conflict topology, and causal trajectory motifs, advancing the vision of trustworthy, explainable GeoAI for public safety.

Acknowledgments

The VTTSI system was developed in collaboration with the Virginia Tech Transportation Institute (VTTI). Generative AI tools, including OpenAI ChatGPT (GPT-4o) and Anthropic Claude, were utilized to assist in generating portions of this work, including manuscript text, $\LaTeX$ formatting, and code snippets. All AI-generated content was reviewed, verified, and edited by the authors, who take full responsibility for the accuracy and integrity of the submitted manuscript.

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